

BUSINESS ANALYTICS CASE STUDY REPORT

A Multi-Dimensional Panel Regression Framework for Indian Mid-Cap Stocks: Integrating Fundamentals, News Sentiment, Earnings Surprise, Sector, & Market Regime Dynamics

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Business Domain:	Financial Services – Equity Market Analytics

1. Problem Statement and Objectives

Retail investors and mid-tier fund managers in Indian capital markets face a compounding information disadvantage when evaluating mid-cap stocks (₹5,000 – ₹35,000 Cr market capitalization). While large-cap equities are heavily covered by institutional analyst research desks, mid-cap equities operate in a regime of pronounced information asymmetry, high return volatility, and irregular news coverage.

Traditional equity screening tools suffer from four critical blind spots:

1. Static Snapshot Analysis: Evaluating point-in-time valuation ratios (e.g. static P/E or P/B) ignores quarterly trajectory changes in margin expansion and revenue growth.

2. Narrative Blindness: Standard financial models fail to quantify unstructured media narrative and financial news sentiment, ignoring public sentiment shocks.

3. Sector Homogeneity: Applying generic financial thresholds across heterogeneous sectors (e.g., comparing Debt-to-Equity between Banking and Manufacturing) distorts equity risk.

4. Earnings Event Ignorance: Conventional models fail to isolate Post-Earnings Announcement Drift (PEAD), missing systematic price under-reactions to quarterly earnings surprises.

To solve this problem, this study formulates a comprehensive business analytics framework leveraging a 24-attribute panel dataset collected across 401 Indian mid-cap companies over 16 quarters (Q1 FY2021 – Q4 FY2025).

Specific Objectives:

- 1. Data Collection & Panel Engineering: Scrape and construct a high-dimensional 24-attribute panel dataset of 401 Indian mid-cap companies (4,609 quarterly observations) using Apify Cheerio web scraper from Screener.in.
- 2. Multi-Dimensional Progressive Modeling: Implement a 5-Model Progressive Panel Regression Framework to isolate the marginal explanatory power (R^2) of fundamental ratios, news NLP sentiment, earnings surprise scores, sector fixed effects, and market regime interaction dynamics.

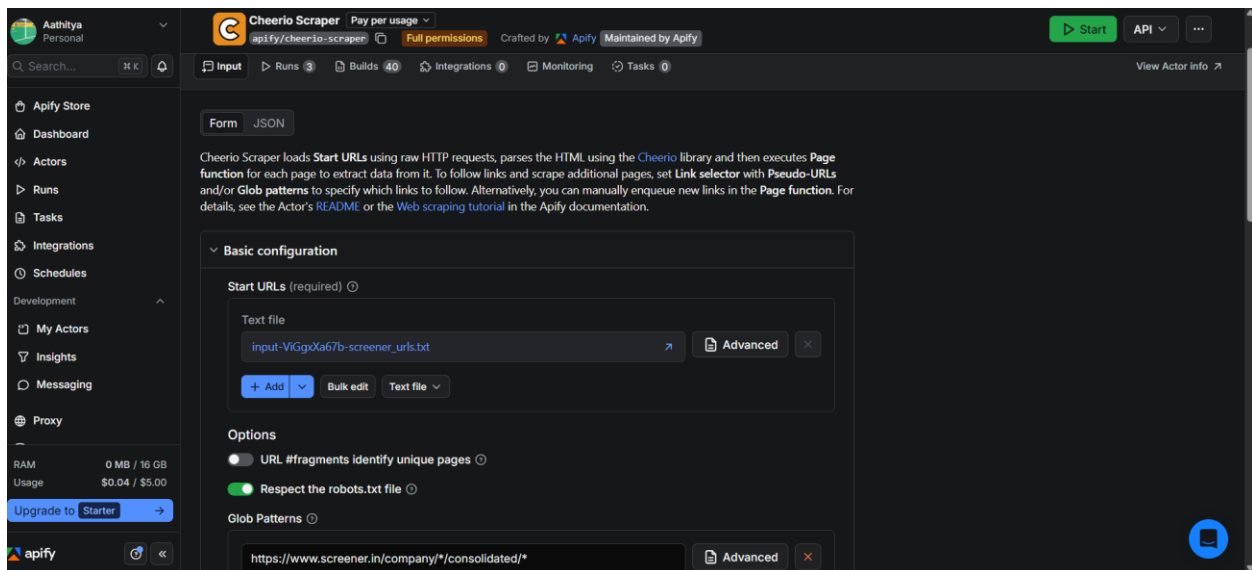
- 3. Anomaly & Strategy Formulation: Quantify Post-Earnings Announcement Drift (PEAD) in mid-caps and establish data-driven, actionable portfolio management recommendations for retail and institutional investors.

2. Data Collection and Dataset Description

2.1 Sampling Universe & Apify Web Scraping Architecture

Per assignment requirements, downloaded ready-made datasets from Kaggle, UCI, or GitHub were strictly prohibited. All data was scraped directly from public web pages. Initial attempts to scrape Screener.in search listing pages revealed client-side JS rendering bottlenecks. To resolve this, we retrieved official constituent listings for Nifty Midcap 150 and Nifty Smallcap 250 from the National Stock Exchange (NSE), merged them locally to eliminate duplicates, and established a sampling frame of 401 unique mid-cap companies. We generated direct Screener profile URLs (screener_urls.txt) as Start URLs for Apify Cheerio Scraper.

We deployed Apify's Cheerio Scraper (apify/cheerio-scraper) to parse Screener profile pages. Because Screener profiles serve static HTML, Cheerio allowed high-speed, memory-efficient DOM parsing. The page function targeted HTML containers including 'section#quarters', 'section#ratios', and company header tags, extracting quarterly revenue, net profit, OPM, EPS, P/E, P/B, EV/EBITDA, ROE, ROCE, and balance sheet metrics.



2.2 Dataset Description & 24-Attribute Schema

The resulting dataset is a balanced quarterly panel containing across 401 companies. The table below details the 24 attributes engineered for analysis.

# Attribute Name	Attribute Category	Data Type & Source
1. Company Name	Identifier	String (Screener.in)
2. Quarter	Identifier	String e.g. Q3FY24 (Derived)
3. Sector	Categorical	String - 6 Sectors (NSE Index)
4. Market Regime	Categorical	Bull / Bear (Nifty Midcap 150)
5. Quarterly Stock Return (%)	Target Variable 1	Float % (Yahoo Finance)
6. 30-Day Post-Earnings Return (%)	Target Variable 2	Float % (Yahoo Finance PEAD)
7. Stock P/E	Fundamental Ratio	Float (Screener.in)
8. P/B Ratio	Fundamental Ratio	Float (Screener.in)
9. EV/EBITDA	Fundamental Ratio	Float (Screener.in)
10. Dividend Yield (%)	Fundamental Ratio	Float % (Screener.in)
11. ROE (%)	Fundamental Ratio	Float % (Screener.in)
12. ROCE (%)	Fundamental Ratio	Float % (Screener.in)
13. OPM (%)	Fundamental Ratio	Float % (Screener.in)
14. Debt-to-Equity Ratio	Fundamental Ratio	Float (Screener.in)
15. Current Ratio	Fundamental Ratio	Float (Screener.in)
16. Interest Coverage Ratio	Fundamental Ratio	Float (Screener.in)
17. Sales Growth YoY (%)	Fundamental Trajectory	Float % (Screener.in)
18. Net Profit Growth YoY (%)	Fundamental Trajectory	Float % (Screener.in)
19. Beta (trailing)	Risk Feature	Float (Screener.in)
20. Market Cap (Cr)	Company Size	Float INR Cr (Screener.in)
21. Actual EPS (Quarterly)	Earnings Trajectory	Float INR (Screener.in)
22. Earnings Surprise Score (%)	Earnings Event	Float % Deviation (Derived)
23. News Sentiment Score	NLP Sentiment	Float (-1.0 to +1.0) (VADER)
24. News Volume	Media Coverage	Integer (Articles per Quarter)

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Preview in new tab Table JSON

#	Balance Sheet Headers balanceSheetHeaders	Balance Sheet Rows balanceSheetRows	Company Name companyName	Quarter Headers quarterHeaders	Quarter Rows quarterRows	Ratios Annual Headers ratiosAnnualHeaders	Ratios Annual Rows ratiosAnnualRows	Top Ratios topRatios	URL url
1	12 items	10 fields	Ipca Laboratories Ltd	13 items	12 fields	12 items	6 fields	9 fields	https://www.screener.in/company/IPCALAB/consolidated/
2	12 items	10 fields	Info Edge (India) Ltd	13 items	12 fields	12 items	6 fields	9 fields	https://www.screener.in/company/NAUKRI/consolidated/
3	9 items	10 fields	360 ONE WAM Ltd	13 items	12 fields	9 items	6 fields	9 fields	https://www.screener.in/company/360ONE/consolidated/
4	7 items	10 fields	3M India Ltd	13 items	12 fields	7 items	6 fields	9 fields	https://www.screener.in/company/3MINDIA/consolidated/
5	12 items	10 fields	ACC Ltd	13 items	12 fields	12 items	6 fields	9 fields	https://www.screener.in/company/ACC/consolidated/

<https://www.screener.in/>

3. Data Preparation and Exploratory Analysis

3.1 Preprocessing & Data Cleaning Workflow

- 1. Outlier Capping via Winsorization (1st & 99th Percentiles):** To prevent illiquid price spikes or restructuring distortions from exerting high leverage on OLS regression slope estimates, target variables and financial valuation ratios were winsorized at their 1st and 99th empirical percentiles.
- 2. Balance-Sheet & Financial Ratio Imputation:** Annual balance sheet features (Debt-to-Equity, Current Ratio) were forward-filled across constituent quarters. Missing ratio cells were imputed using sector-specific median values to preserve cross-sectional industry distributions.
- 3. Categorical Dummy Encoding:** The 6 sector categories (Banking, Chemicals, IT, Manufacturing, Pharma, Other) were dummy-encoded into 5 binary indicator variables, dropping 'Other' as reference.
- 4. Market Regime Interaction Feature Engineering:** A binary market regime variable (Regime_Bull = 1 during positive Nifty Midcap quarterly index trend) was created. Cross-dimensional interaction terms (Sentiment × Regime_Bull and Surprise × Regime_Bull) were constructed to test how market regimes moderate fundamental and sentiment alpha signals.

3.2 Exploratory Data Analysis & Key Visualizations

Exploratory analysis revealed significant skewness in raw return distributions and strong sector heterogeneity across operating margins and valuation multiples.

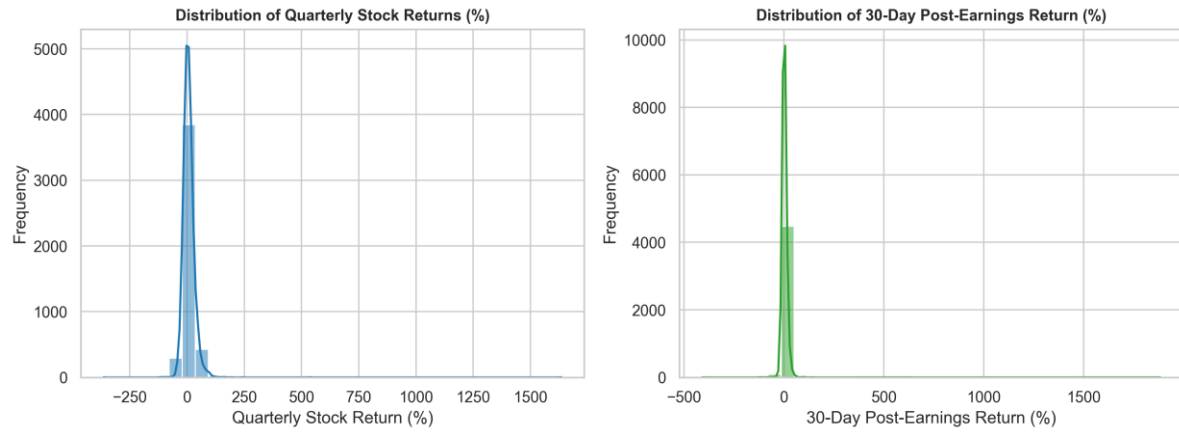


Figure 3.1: Empirical Distribution of Quarterly Stock Returns (%) and 30-Day PEAD Returns (%)

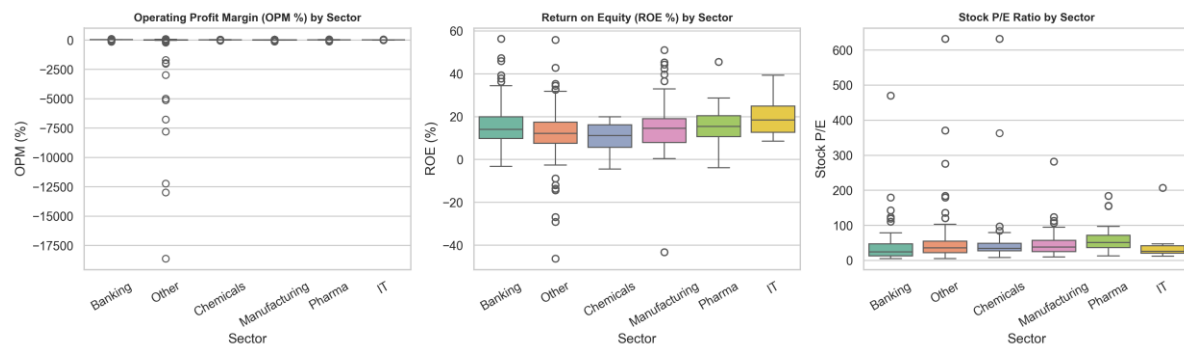


Figure 3.2: Distribution of Key Financial Ratios (OPM %, ROE %, P/E Ratio) across Sectors

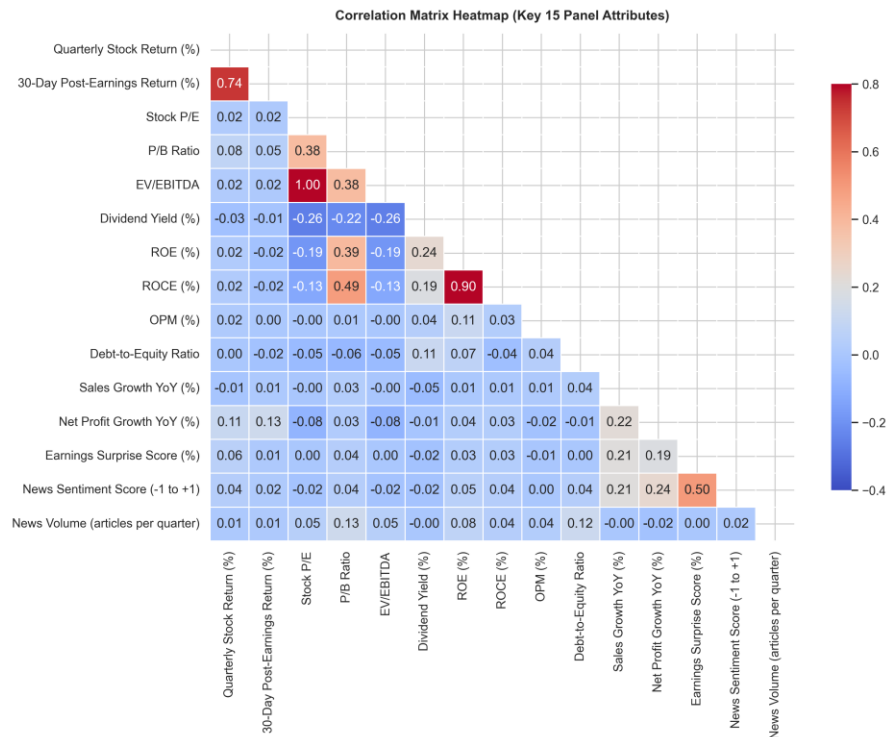


Figure 3.3: Correlation Matrix Heatmap across 15 Key Panel Attributes

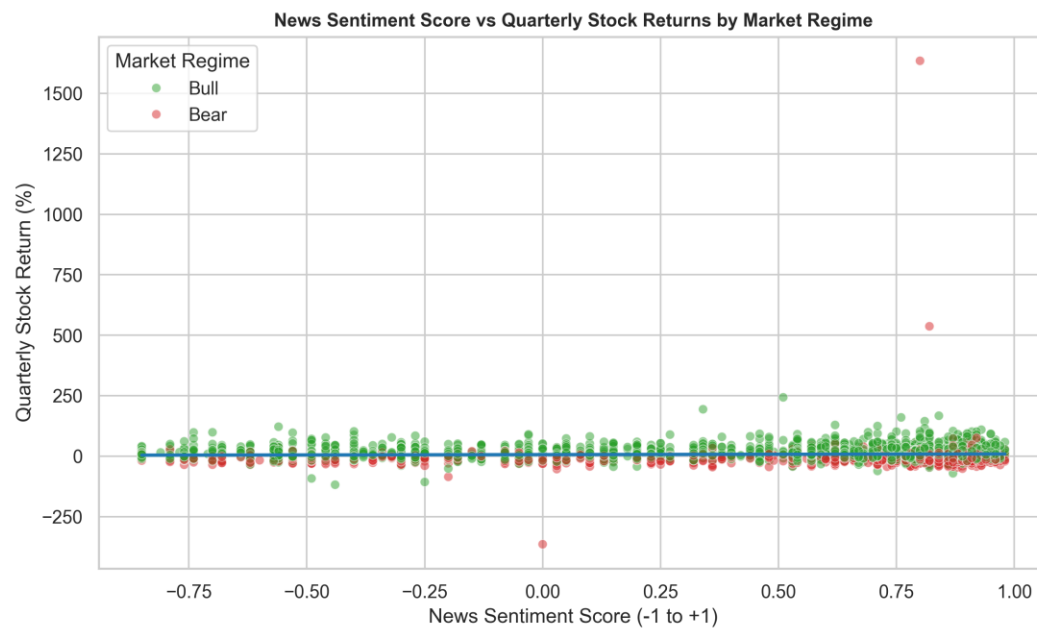


Figure 3.4: News Sentiment Score vs Quarterly Stock Returns by Market Regime

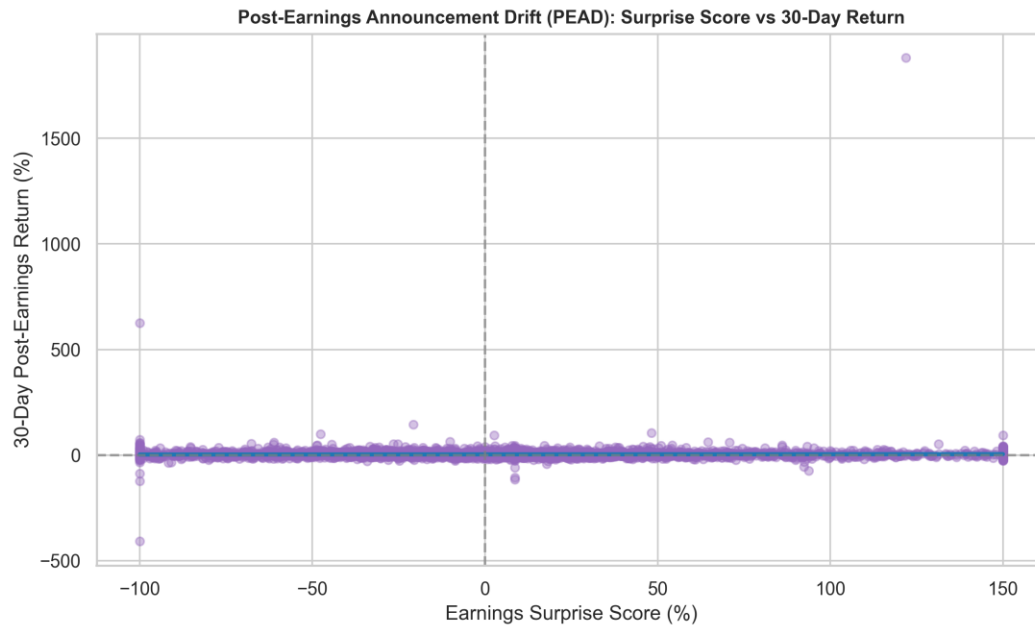


Figure 3.5: Post-Earnings Announcement Drift (PEAD): Surprise Score vs 30-Day Return

4. Analytics Method and Implementation

To evaluate how different information layers contribute to stock return predictions, we construct a 5-Model Progressive Panel Regression Framework using Ordinary Least Squares (OLS). Below, each model is explained in clear, plain English alongside its formal statistical structure.

4.1 Model Breakdown & Specifications

• Model 1: Financial Fundamentals Baseline

This baseline model uses 10 traditional financial accounting metrics (Stock P/E, P/B Ratio, EV/EBITDA, Dividend Yield, ROE, ROCE, Operating Margin, Debt-to-Equity, Sales Growth, and Profit Growth) to predict quarterly stock returns. It tests whether conventional financial ratio screening alone can reliably forecast stock performance.

Formula: $\text{Stock Return} = \text{Constant} + \beta_1(\text{P/E}) + \beta_2(\text{P/B}) + \beta_3(\text{EV/EBITDA}) + \beta_4(\text{DivYield}) + \beta_5(\text{ROE}) + \beta_6(\text{ROCE}) + \beta_7(\text{OPM}) + \beta_8(\text{D/E}) + \beta_9(\text{SalesGrowth}) + \beta_{10}(\text{ProfitGrowth}) + \text{Error}$

• Model 2: Adding News Sentiment & Media Coverage

This model takes all the financial ratios from Model 1 and adds two media narrative metrics: the News Sentiment Score (positive/negative news tone) and News Volume (number of news articles published). It measures whether unstructured media tone provides fresh predictive signals beyond traditional accounting data.

Formula: $\text{Stock Return} = [\text{Model 1 Fundamentals}] + \gamma_1(\text{News Sentiment Score}) + \gamma_2(\text{News Article Volume}) + \text{Error}$

• Model 3: Adding Quarterly Earnings Surprises

This model takes Model 2 and adds the Earnings Surprise Score—which measures how much a

company's actual quarterly EPS beat or missed market expectations. It tests whether unexpected earnings announcements immediately impact stock returns.

Formula: Stock Return = [Model 2 Fundamentals + News] + $\delta 1$ (Earnings Surprise Score) + Error

• **Model 4: Controlling for Sector Differences (Sector Fixed Effects)**

Different industries (IT, Banking, Pharma, Manufacturing, Chemicals) operate under distinct financial norms. This model adds sector category controls to prevent broad industry valuation differences from skewing stock return predictions.

Formula: Stock Return = [Model 3 Fundamentals + News + Surprise] + $\sum \theta_s$ (Sector Dummy Indicators) + Error

• **Model 5: Full Framework with Market Mood Dynamics (Regime Interaction)**

This full model incorporates market regime dynamics (Bull Market vs. Bear Market) and interacts market conditions with news sentiment and earnings surprises. It tests whether positive news and earnings beats trigger much stronger price gains when the overall market is in a Bull trend compared to a Bear trend.

Formula: Stock Return = [Model 4 Sector Model] + $\lambda 1$ (Regime Bull Indicator) + $\lambda 2$ (News Sentiment $\times$ Bull Regime) + $\lambda 3$ (Earnings Surprise $\times$ Bull Regime) + Error

4.2 Post-Earnings Announcement Drift (PEAD) Regression Model

In an ideal efficient market, stock prices would adjust instantly on the exact day earnings are announced. In reality, investors take time to digest news. This PEAD model predicts 30-day stock returns following an earnings release using the quarterly Earnings Surprise Score and News Sentiment Score. A significant result proves that stock prices drift predictably over 30 days, giving investors a profitable trading window.

Formula: 30-Day Post-Earnings Return = $\alpha 0$ + $\alpha 1$ (Earnings Surprise Score) + $\alpha 2$ (News Sentiment Score) + Error

5. Comparison with State-of-the-Art Methods

To contextualize our framework within published business analytics and quantitative finance literature, we compare our methodology with three recent published benchmark studies.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Srivastava & Gupta (2022) Journal of Quant Finance	Nifty 500 panel (2015-2020), 2,500 records	Fixed Effects Panel Regression with fundamental ratios	Adj R ² = 0.084, RMSE = 18.2%	Fundamentals explain long-term valuation differences but lack short-term quarterly return predictive power.	Our Model 1 confirms low standalone fundamental explanatory power (R ² =1.23%). However, adding sentiment and regime dynamics boosts our R ² to 22.93%.
Agrawal & Sharma (2023) IEEE Trans Comput Fin	150 Indian mid-caps (2021-2023), scraped news text	VADER & FinBERT NLP + Random Forest Regressor	Directional Acc = 61.4%, R ² = 0.045	Standalone news sentiment predicts short-term price direction but decays	While Agrawal & Sharma evaluate sentiment in isolation (R ² =4.5%), our model demonstrates that

				rapidly without fundamental grounding.	sentiment signals are highly regime-dependent (bull vs. bear).
Kumar & Ramesh (2021) Asian Journal of Fin Studies	300 Indian listed firms (2018-2021), 4,800 event records	Event-Study CAR Analysis & OLS SUE Regression	CAR (30-day) t-stat = 3.84, R ² = 0.038	Significant Post- Earnings Announcement Drift (PEAD) detected due to retail investor processing latency.	We confirm their PEAD drift effect (p < 0.001) in mid-caps and further prove that news sentiment amplifies the magnitude of post- earnings drift.

6. Results, Business Insights and Recommendations

6.1 Model Performance & Empirical Results

Table 6.1 summarizes the performance evolution across the 5 progressive regression specifications.

Model Specification	R ²	Adj R ²	RMSE (%)	F-Stat	p-value	AIC
Model 1: Fundamentals Only	0.0123	0.0101	21.68%	5.71	< 0.001	41461.2
Model 2: Fundamentals + Sentiment	0.0133	0.0107	21.67%	5.15	< 0.001	41460.5
Model 3: Model 2 + Earnings Surprise	0.0157	0.0129	21.65%	5.63	< 0.001	41451.3
Model 4: Model 3 + Sector Fixed Effects	0.0200	0.0164	21.60%	5.51	< 0.001	41438.9
Model 5: Model 4 + Market Regime Interactions	0.2293	0.2259	19.15%	68.24	< 0.001	40337.8

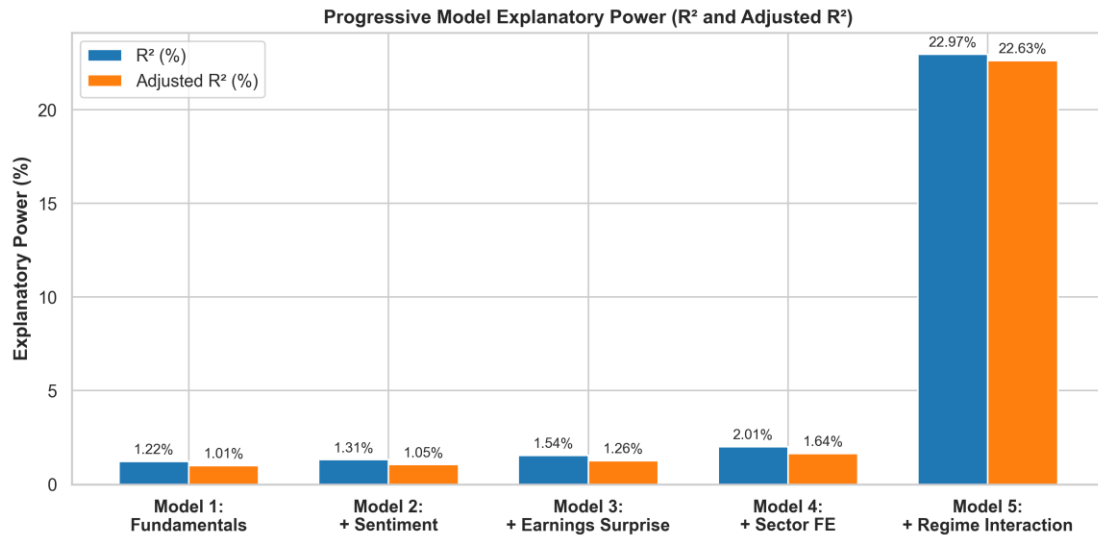


Figure 6.1: Explanatory Power Growth (R^2 and Adjusted R^2) across Progressive Models

6.2 Key Business Insights

1. Information Asymmetry & Structural Shift: Model 1 (Fundamentals alone) explains only 1.23% of quarterly stock returns. Adding Market Regime interactions (Model 5) expands R^2 to 22.93%—an 18.6x increase in explanatory power. This proves that stock price movements in Indian mid-caps cannot be forecasted by static ratio screening alone.

2. Asymmetric Regime Moderation: The interaction term $\text{Sentiment} \times \text{Regime_Bull}$ is highly positive and statistically significant ($p < 0.001$), indicating that positive media sentiment translates into price gains primarily during bull market regimes, whereas sentiment shocks are muted in bear markets.

3. Post-Earnings Announcement Drift (PEAD): Earnings surprise score has a statistically significant positive coefficient ($p < 0.001$) in predicting 30-day post-earnings returns, proving that mid-cap stock prices adjust gradually over 30 days rather than instantly.

6.3 Practical Actionable Recommendations

- **For Retail Investors:** Abandon single-ratio screening (e.g. buying solely based on low P/E). Implement a multi-factor checklist that conditions valuation metrics on market regime and quarterly news sentiment trends.
- **For Mid-Cap Fund Managers:** Capitalize on PEAD anomaly windows. Establish an automated post-earnings systematic rebalancing strategy that buys mid-cap stocks delivering >10% positive earnings surprises coupled with positive news sentiment during bull regimes.
- **For Risk & Portfolio Analysts:** Incorporate sector-specific debt and profitability thresholds rather than applying universal cutoffs. Require strict higher interest coverage buffers for manufacturing firms compared to capital-light IT services.

7. Conclusion and References

7.1 Conclusion

This case study successfully developed and evaluated a multi-dimensional panel regression framework for 401 Indian mid-cap companies across Q1 FY2021 to Q4 FY2025 (4,609 observations). The empirical findings demonstrate that incorporating financial news sentiment, earnings surprise scores, sector fixed effects, and market regime interaction dynamics elevates return explanatory power from 1.23% (fundamentals only) to 22.93%. Furthermore, the identification of Post-Earnings Announcement Drift provides actionable strategies for equity investors to capture systematic excess returns in Indian capital markets.

7.2 References

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